A ONE-DIMENSIONAL SEPARATED FINITE TYPE SCHEME ADMITS A SMOOTH PROJECTIVE COMPLETION AFTER A FINITE SEPARABLE EXTENSION OF THE BASE FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement contains a commutative diagram typeset with Xy-pic; the retained chapter preamble supplies the native Xy-pic setup, so it compiles exactly as in the source.

Editorial note. The counted span cites seven cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The author's parenthetical 'details omitted' concerns only the routine transfer from the completion to X; the rest of the argument is written out.

Allowed standard prerequisites (editorial).

- Normal and regular agree in dimension at most one. Stacks Project, Properties, Lemma properties-lemma-normal-dimension-1-regular (tag 0B3B). Turns normality of the normalization into geometric regularity and hence smoothness.
- Fields of global sections of a proper geometrically reduced scheme. Stacks Project, Varieties, Lemma lemma-proper-geometrically-reduced-global-sections (tag 0CK4). Shows that the fields k'_i are finite separable extensions of k'.
- Stein factorization for finite morphisms. Stacks Project, Varieties, Lemma lemma-baby-stein (tag 0C3C). Produces the final commutative diagram with the stated properties.

Source reference. Original lemma 0BXW: sources/stacks-chow/tags.txt, source lines 14994–14994. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0BXT: sources/stacks-chow/tags.txt, source lines 14991–14991. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original section 0BXQ: sources/stacks-chow/tags.txt, source lines 14988–14988. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0BX2: sources/stacks-chow/tags.txt, source lines 14965–14965. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 038X: sources/stacks-chow/tags.txt, source lines 4252–4252. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0BUG: sources/stacks-chow/tags.txt, source lines 14874–14874. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0FD1: sources/stacks-chow/tags.txt, source lines 19201–19201. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a separated scheme of finite type over k with $\dim(X) \leq 1$. Then there exists a commutative diagram*

$$\begin{array}{ccccccc}
\overline{Y}_1 \amalg \ldots \amalg \overline{Y}_n & \xleftarrow{j} & Y_1 \amalg \ldots \amalg Y_n & \xrightarrow{\nu} & X_{k'} & \longrightarrow & X \\
 & \searrow & \downarrow & & \downarrow & & \downarrow f \\
 & & \mathrm{Spec}(k'_1) \amalg \ldots \amalg \mathrm{Spec}(k'_n) & \longrightarrow & \mathrm{Spec}(k') & \longrightarrow & \mathrm{Spec}(k)
\end{array}$$

of schemes with the following properties:

(1) *k'/k is a finite purely inseparable extension of fields,*
(2) *ν is the normalization of $X_{k'}$,*
(3) *j is an open immersion with dense image,*
(4) *k'_i/k' is a finite separable extension for $i = 1, \ldots, n$,*
(5) *$\overline{Y}_i$ is smooth, projective, geometrically irreducible dimension ≤ 1 over k'_i.*

Proof. As we may replace X by its reduction, we may and do assume X is reduced. Choose $X \to \overline{X}$ as in Lemma 0BXW. If we can show the lemma for $\overline{X}$, then the lemma follows for X (details omitted). Thus we may and do assume X is projective.

Choose k'/k finite purely inseparable such that the normalization of $X_{k'}$ is geometrically normal over k', see Lemma 0BXT. Denote $Y = (X_{k'})^\nu$ the normalization; for properties of the normalization, see Section 0BXQ. Then Y is geometrically regular as normal and regular are the same in dimension ≤ 1, see Properties, Lemma 0BX2. Hence Y is smooth over k' by Lemma 038X. Let $Y = Y_1 \amalg \ldots \amalg Y_n$ be the decomposition of Y into irreducible components. Set $k'_i = \Gamma(Y_i, \mathcal{O}_{Y_i})$. These are finite separable extensions of k' by Lemma 0BUG. The proof is finished by Lemma 0FD1. □